INVESTMENT & POLICY CONCEPT · INDIA · 2026

The Grain Value Loop

An integrated rice biorefinery model that turns the byproducts India's government already procures — and currently gives away — into edible oil, food-grade derivatives, energy, and exports. Feedstock the state owns; demand the state can contract.

IMPORT SUBSTITUTION	BRAN GIVEN AWAY TODAY	CAPITAL SUBSIDY
₹21,000 cr	**₹7,650 cr**	**35–50%**
per year, across crops	in free CMR byproduct	PMFME · PMKSY · PLISFPI

FCI-warehouse-MSPvsMSV · figures verified against DFPD, DGCI&S, MoFPI, real project DPRs

THE PARADOX

India imports what it grows — and gifts the feedstock

The government procures ~550 LMT of rice and ~300 LMT of wheat a year, then hands the milling byproducts to private millers for free and imports the finished derivative.

WHAT THE STATE DOES TODAY

- Procures paddy at MSP, hands it to CMR millers

WHAT THE STATE IMPORTS

- **57% of edible oil** (~₹1.68 lakh cr/yr) — while #1 in rice-bran oil

- Millers keep **~180 LMT husk + ~65 LMT bran** — the ₹7,650 cr byproduct giveaway
- Sells surplus grain raw (OMSS) at ~2–4% trader margin, often at a loss

- **67 LMT of pulses**, vital wheat gluten, specialty starch, Vitamin-E, silica
- The *derivative* — while holding the raw material

> World's #1 rice-bran-oil producer, yet imports more than half its cooking oil. The same pattern repeats for gluten, starch, protein, and nutraceuticals across every crop the state procures.

Sources: DFPD Annual Report 2024–25 · DGCI&S FY26 · NITI Aayog edible-oil

THE INSIGHT

Value density rises ~1,000× from grain to derivative

The margin is in the cascade, not the commodity — the sugarcane model (sugar + ethanol + power + CBG) ported to grain.

Paddy	→	Milled rice + bran + husk	→	Rice-bran oil	→	Oryzanol / Vit-E
₹24/kg (MSP)		the split the state gives away		₹105/kg		₹1,200–17,700/kg

COMMODITY FLOOR	INDUSTRIAL MID	MOLECULAR TAIL
₹7–40/kg	**₹40–110/kg**	**₹350–17,700/kg**
flour, feed, husk, power — where the state operates now	RBO, silica, starch, gluten, dal, besan	wax, protein, oryzanol, tocotrienols

The state captures the bottom tier and gifts the top three. Source: byproduct product-tree (39 products)

THE PROJECT

An integrated rice biorefinery — one complex, many streams

A cooperative/PPP-hosted complex, sited in an operational Mega Food Park, that takes government CMR paddy and runs the full cascade instead of milling and selling raw:

Rice mill · Bran stabilisation · RBO solvent extraction · RBO refinery · Oryzanol / wax recovery · Husk-fired power · Husk-ash → silica · Broken rice → poha/murmura · Fortified rice kernels · DORB → feed

WHY INTEGRATED

- **Husk powers the process** (already standard mill practice) — energy self-sufficient
- **DORB feed credit** makes the oil cheap (₹60 vs ₹105/kg)
- One feedstock stream → 8+ revenue lines
- The distillery/mill is the natural aggregation host

> Edible oil first (import substitution at ₹105/kg), fuel only from the non-edible residual — India must diverge from the US "distillers-oil-to-biodiesel" model.

Grounded in NCDC cooperative rice-mill DPR + SEA (30 MT existing crush capacity)

RAW MATERIAL · PRICES & SOURCES

Feedstock the state already owns

INPUT	INDICATIVE PRICE	SOURCE / BASIS

Paddy (common)	₹2,369/qtl	MSP KMS 2025-26 · govt CMR procurement (~832 LMT)
Rice bran	₹2,000/qtl (₹20/kg)	NCDC DPR · to oil extractors; 16–22% oil
Rice husk	₹2,000/MT	NCDC DPR · fuel-grade; ~20% of paddy
Broken rice	₹1,500/qtl	NCDC DPR · poha/starch/ethanol feedstock
De-oiled bran (DORB)	₹15/kg (credit)	feed byproduct — the credit that makes RBO viable

AGGREGATION MODEL

PACS / FPO / cooperative societies — *already the paddy-procurement centres* — aggregate paddy + bran. The feedstock chain exists; it is currently dispersed to 100,000+ private mills.

SCALE AVAILABLE TO GOVERNMENT

~832 LMT CMR paddy → ~40 LMT government-touchable bran (41% of national) → ~6.4 LMT RBO at full capture.

Prices: NCDC "Establishment of Rice Mill" DPR (Chhattisgarh) · DFPD MSP · vintage noted

UNIT ECONOMICS · FROM REAL DPRS

Margin and IRR — ranked, government-DPR-backed

PRODUCT	NET MARGIN	IRR	DPR SOURCE
Poha / murmura	10–15%	<3y payback	NIFTEM-T PMFME ✓
RBO solvent extraction	8–15%	–	IMARC/EIRI
Rice mill (anchor)	8.5–15%	19%	NCDC ✓
Maize wet-milling (starch)	12–20%	22–28%	industry
Dal mill	~5.3%	25–31%	RACP + NIFTEM ✓
Roller flour mill	3–5.5%	20–29%	RACP ✓

Margin ≠ return. The staples show thin 3–5% margins but **19–31% IRR** — low capex, fast turnover. The "process, don't sell raw" case rests on return-on-capital, not margin-on-sales.

Sources: NCDC · Rajasthan RACP (Grant Thornton) · NIFTEM–Thanjavur PMFME · IMARC — verified/reported labelled

SCHEMES · THE FUNDING STACK

35–50% of capital comes from MoFPI

PMFME

35%

credit-linked capital subsidy, max ₹10 L/unit · **micro** units (RBO, dal, poha, besan)

PMKSY

35–50%

grants for clusters, cold chain, agro-processing (50% NE/difficult) · **the integrated complex**

PLISFPI

6-yr

incentive on incremental sales + branding-abroad · **large branded** (RBO brand, exports)

MEGA FOOD PARK INFRASTRUCTURE

23 operational parks (of 41), ~₹120 cr each with ~₹50 cr PMKSY grant — ready common facilities (power, cold chain, warehousing, effluent) to plug into, not build.

THE SUBSIDY IS A MARGIN MULTIPLIER

Anchor DPR: ₹80 L cost → ₹16 L subsidy + ₹52 L loan + ₹12 L own. The 19% project IRR becomes a much higher return on the ₹12 L actually invested. Capital-heavy units (RBO, silica, wet-milling) gain the most from a 35–50% grant.

Sources: pmfme.mofpi.gov.in · PMKSY FAQs · mofpi.gov.in/PLISFPI · MoFPI 41-MFP list (31.01.2023)

MARKETS · THREE DEMAND TIERS

Demand the government can contract

CAPTIVE · SUREST

- **Fortified rice kernels — ₹17,082 cr program** to 2028, 100% GoI-funded; ~3.5–4 LMT/yr standing demand
- Fortified atta · Vitamin-E fortificant from own bran

DOMESTIC GROWTH

- **RBO** — India #1 market, $1.39 bn; +exports $104.5 mn
- Poha/murmura (snacks 14.5% CAGR) · besan
- Maize starch $4.1 bn · wheat gluten

EXPORT-LED

- **Basmati ₹50,000 cr** (FY25, +15.7%, ₹82/kg)
- Plant protein (14% CAGR) · spices ₹39,994 cr · green silica (Tata ₹775 cr)

> The offtake is **contractable, not consumer-dependent**: food-processing supply agreements + fortification mandates + institutional (PDS/ICDS) offtake form a demand backbone independent of the retail shelf — the ethanol-blending-mandate lever, applied to processed grain.

Sources: APEDA FY25 (basmati ₹50k cr) · DFPD FRK program · DGCI&S · industry market reports — verified/reported labelled

THE PRIZE · IMPORT SUBSTITUTION

₹21,000 cr/yr across the crops the state holds

STREAM	IMPORT ₹/T (DGCI&S)	SAVINGS ₹ CR/YR	OFFTAKE
Rice bran oil	97,970	7,544	frying/HoReCa + fortification
Pulses (dal margin)	50,888–74,163	~7,000	dal millers + PDS
Cottonseed oil	98,766	1,975	food processing
Maize starch	37,170	1,115	food/pharma/paper

Husk → silica	86,730	867	tyre/industrial
Protein · gluten · Vit-E · lecithin	115k–17.7m	~1,900	processors + export

HIGH-CONFIDENCE TIERS

₹18–21k cr

oil + pulses + industrial mid

ON FEEDSTOCK THE STATE

already owns

via CMR procurement

THROUGH OFFTAKE IT CAN

contract

not hope for

Source: consolidated-savings-by-crop · DGCI&S FY26 import unit values · illustrative, additive

A CONCRETE MSV CASE · WHEAT → PASTA

Durum wheat: the fastest-growing value-add India under-grows

The same "import the derivative, hold the feedstock" pattern — in wheat. India is **wheat-surplus** and sells it cheap via OMSS (~₹2,550/qtl), yet:

- Pasta market **$1.21 bn, growing ~14–16%/yr** — the fastest wheat-product segment
- India is the **world's #2 durum-semolina importer** — pasta needs durum (hard) wheat, which India barely grows (MP's Malwa belt, ~1.5–2 MT)
- DGCI&S: pasta imports **+24%**, biscuit imports **+28%**, gluten **+35%** — the value-added tier's imports are building

THE DOUBLE PLAY (DURUM MSV)

- **Procure durum by variety** — a variety-specific MSV that de-risks durum cultivation (the wheat parallel to the oilseed-MSP production lever)
- **Mill semolina → pasta** on OMSS/durum wheat, in a Mega Food Park
- Substitutes **both** the pasta imports AND the durum-wheat imports

- **Export prize:** India ships only **$63 mn** of pasta into a global import trade of **US $1.62 bn · Germany $1.24 bn** — under 4% of the US market, on cheap OMSS wheat Italy can't match

- + vital gluten (13,000 t/yr imported) from the same wheat wet-mill

> India is a net *exporter* of biscuits (Parle-G, 180 countries) — so the target is the **premium/pasta/durum tier** where imports are growing, not the mass market. Import substitution is already visible: durum-pasta imports fell −25% YoY.

Sources: DGCI&S TradeStat EIDB (exact 8-digit HS, FY25–26 prov.) · industry market sizes · report/wheat-value-added-products.md

A CONCRETE CASE · EDIBLE OIL FROM FCI/CMR STOCK REUTILISATION

The oil India imports for ₹1.68 lakh cr — from grain the state already holds

India imports **168 LMT of edible oil — US$19.3 bn, ₹1.68 lakh cr** a year (57% of demand), landing at **~₹100/kg crude** (palm ₹96). Yet the 24 MSP crops **are** the domestic oil-feedstock base — and FCI's own procured paddy throws off the cheapest oil in the country as a byproduct:

- **Rice bran oil** — bran from the ~130 MT of paddy under CMR; **₹60/kg ex-mill vs ₹105 import parity = ₹45/kg edge**. India is the world's #1 RBO producer — and hands the bran to millers **free** (the ₹7,650 cr giveaway)
- RBO is also the *better* oil: γ-oryzanol, tocotrienols, phytosterols; cholesterol-

IMPORT-SUBSTITUTION SAVINGS ESTIMATE

STREAM	₹/KG EDGE	₹ CR/YR
Rice bran oil (7.7 LMT)	45	7,544
Cottonseed oil (2 LMT)	~10	1,975
Soy/sun crush — cake + forex*	system	~1,500
Capturable now		**~₹11,000 cr/yr**

*at-parity oils save via the oil-cake co-product imports don't carry + forex retained, not a retail-price cut.

lowering; high-heat stable; non-GMO (ICRBO-2016)

- **Cottonseed oil** — byproduct of cotton (an MSP crop): crush margin + cake

> Honest tiering: **~₹11,000 cr/yr is capturable now** from byproduct oils on feedstock the state already procures. The **production lever** (NMEO-OS oilseeds 39→69.7 MT) is the ceiling — closing even half the 168-LMT gap replaces **~₹84,000 cr** of imports over 5–7 yrs — but mustard/groundnut are premium oils, a *volume/self-sufficiency* lever, never an in-year cheaper-cooking-oil promise.

Sources: DGCI&S TradeStat EIDB (168 LMT/₹1.68 L cr) · DFPD edible-oil scenario · SEA (16.3 MT/₹1.61 L cr) · consolidated-savings-by-crop · report/msp-crops-refined-oil-source.md

THE PROOF IT WORKS · A MANDATE ALREADY SAVED ₹1.36 LAKH CR

Ethanol proves the model · OMSS the channel · rice-bran oil the un-pulled lever

1 · ETHANOL — the model
38 → **661 cr L**; blending **20% by Mar-2025** (5 yrs early), on broken rice + maize. A blending *mandate* built the offtake.
₹1.36 lakh cr forex saved

→

2 · OMSS — the channel
FCI already sells surplus grain to millers at reserve price: **71 LMT wheat + 1.6 LMT rice** (2023-24). OMSS rice → millers → **bran**.
the RBO feedstock pathway

→

3 · RBO — the lever
India is **#1 in rice-bran oil** (NITI, Aug-2024), yet leaves the bran's oil in cattle feed. **₹60 vs ₹105/kg** import parity = ₹45/kg edge.
the un-pulled lever

GOVT CMR BRAN (~40 LMT)
₹6,698 cr
6.4 LMT RBO — import displaced, feedstock the

ALL-INDIA BRAN (~98 LMT)
₹16,464 cr
15.7 LMT RBO — the ceiling

REALISTIC NEAR-TERM (+7.7 LMT)
₹8,064 cr
~5% of the ₹1.68 lakh-cr oil bill; +₹5,616 cr DORB

state controls		feed

> The ethanol mandate turned surplus/broken grain into **₹1.36 lakh cr of forex saved**. The edible-oil bill (₹1.68 lakh cr) is the same-size prize — and **RBO is the grain-side lever not yet pulled**: bran the state already owns and gives away free, with a PDS-oil / fortification mandate as the offtake, just as blending was for fuel.

`Sources: PIB PRID 2113234 (ethanol) · PIB PRID 2200287 & NITI Aayog Aug-2024 (India #1 in RBO) · PIB PRID 2000937 (OMSS 2023-24) · rbo_savings_scenarios.py · report/omss-ethanol-rbo-import-substitution.md`

ANSWERING THE OBJECTION · FUEL VS FOOD SECURITY

Byproduct oil isn't fuel-vs-food — it competes with imports, not the plate

The debate only exists when you burn the *grain*. It vanishes the moment you use the **byproduct**.

ROUTE	FEEDSTOCK	COMPETES WITH FOOD?
Grain → ethanol	broken/surplus grain	partially — the debate lives here
Bran → RBO	milling byproduct	**No** — rice still feeds people
Cottonseed → oil	cotton byproduct	**No**
Husk → silica/energy	byproduct	**No**

Paddy is milled and the **rice feeds people through PDS exactly as before** — only the bran (today given to cattle with its oil intact) is processed. No grain leaves the food chain.

A CLEAN DECISION RULE

- **Byproducts first** — bran/cottonseed/husk: zero food tradeoff, pure import substitution, unlimited go-ahead
- **Grain-to-fuel only above a stock threshold** — institutionalise the **July-2023 FCI-rice-to-ethanol pause** (food already outranks fuel when stocks tighten)
- **Never compress the entitlement** or divert issue-grade grain — the 80-cr guarantee is ring-fenced
- **Least water-intensive feedstock** — maize/byproducts over paddy ethanol in stressed basins

Spoilage is negligible (0.002–0.022% of off-take) — the grain is fine; the question is its *best use*.

> The real choice isn't food vs fuel — it's whether to **capture the value of byproducts India already generates and gives away, or keep importing the derivative**. RBO substitutes ₹1.68 lakh cr of edible-oil *imports*, not the poor's ration.

Sources: PIB (ethanol/EBP; July-2023 FCI-rice pause) · DFPD Foodgrain Bulletin (off-take/spoilage) · report/fuel-vs-food-security.md

THE LIVE INSTRUMENT · BPCL/OMC 40% FCI-RICE ETHANOL CLAUSE (ESY 2025-26)

From 440 lakh ha of rice to a 52-LMT ethanol slice — is there "sufficient grain"? Yes.

RICE FLOW — WHERE THE GRAIN GOES (LMT · DA&FW / DFPD)

Production 1,378
cultivation · ~440 L ha

FCI proc. 520
38% of crop (rest = private/consumption/seed/export)

PDS/schemes 400
NFSA + PM-POSHAN + ICDS off-take

Eth. 52
ethanol segment — ~234 cr L (~3.8% of crop)

CARRY-OVER (RICE OPENING STOCK, 1 JUL) VS NORM ~135 LMT

2021	2022	2023	2024
297	317	253	326

Stock runs **2–2.4× the buffer norm** every year — a persistent **~150–190 LMT surplus**. The 52-LMT ethanol draw sits comfortably inside it.

THE 40% CLAUSE — EVALUATED

New in ESY 2025-26: FCI surplus rice must be **≥40% of grain-based ethanol** (A ≥ 40%×(A+B+C)). "Sufficient grain" is enforced by **a 52-LMT cap, 30-Jun-26 sunset, and Q4 barred.**

VERDICT

✅ Converts idle surplus → guaranteed offtake	good
✅ Cap + sunset + Q4-bar = stock discipline	right instinct
⚠️ Fixed cap, not a *live* stock trigger	blunt
⚠️ ₹2,320/qtl — **₹5.8/kg below OMSS**	subsidy

Pair it with the byproduct route: the ethanol rice was milled from paddy — the **bran→RBO is captured upstream regardless**, so ethanol + RBO are complementary, not rival.

Fix the clause: make "sufficient grain" a **live stock-threshold trigger** (not a fixed 52-LMT cap), price ethanol-rice at OMSS parity, and mandate that FCI-milled ethanol rice routes its **bran to RBO** — capturing the no-tradeoff edible-oil value on top.

Sources: BPCL/OMC ethanol tender ESY 2025-26 (T-22376) vs 2024-25 · DA&FW Final Est. 2023-24 (rice 1,378 LMT) · DFPD Foodgrain Bulletin (rice stock) · report/ethanol-fci-40pct-clause-evaluation.md

INDEPENDENT VALIDATION · THE STATE'S OWN AUDITOR

The CAG audited the same inefficiency — and put ₹-figures on it

Every efficiency claim here is inferred from public data. The **Comptroller & Auditor General (Report No. 20 of 2023, FY2017-22)** found the same, audited — turning "a plausible reading" into "the supreme auditor agrees."

LTTC FREIGHT REBATE FORGONE

₹1,736 cr

largest single avoidable cost — movement inefficiency (¶4.4.1)

AVOIDABLE HIRING — DESPITE OWN VACANT CAP

₹62.76 cr

+ ₹170 cr avoidable carry-over to SGAs (¶3.4.3 / 3.4.6)

STORAGE AUGMENTATION BUILT

12.25 / 34.5 LMT

only ~36% of the 5-yr target (¶1.1.1)

MAPS TO THE THESIS

- **Over-hiring despite vacant capacity** → size owned/coop to average, hire the peak
- **Sangrur silo — stock blocked** (¶3.6.5) → the idle static base to divert to processing
- **Augmentation under-delivered** → the cooperative/PACS storage plan
- **₹158.88 cr railhead-issue foregone, Jharkhand** (¶4.3.4) → distributed delivery

THE HONEST COUNTERWEIGHT

The CAG critiques **execution, not the outcome**. FCI still runs the world's largest food-security operation with **negligible spoilage (0.002–0.022%)**. Both are true: the entitlement is delivered *and* the storage/movement machinery is materially inefficient — that slack is exactly what value-capture + cooperative storage convert, without touching the ration.

~**₹2,383 cr** of avoidable cost the CAG could quantify (2017-22) — the inefficiency isn't a reason to shrink FCI; it's the **slack a processing + cooperative-storage redesign turns into value.**

Source: CAG Report No. 20 of 2023 — Performance Audit on Storage & Movement of Food grains by FCI · report/cag-audit-fci-findings.md

THE PDS · ONORC CASE · THE DEMAND RAIL ALREADY EXISTS

₹2 lakh cr already spent, 80 cr served — redirect the oil spend to domestic jobs

DOMESTIC JOBS

FAIR PRICE SHOPS

5.43 lakh

99.8% ePoS-automated · dealer + staff livelihoods, every district

RICE MILLS

75k–100k

10–50 workers each · TN 3,399 · AP 1,913 · WB 1,873

SOLVENT-EXTRACTION PLANTS

~350

875 SEA members · 30 MT processing capacity

PRODUCTION VOLUMES

Rice procured	~520 LMT
Wheat procured	300 LMT
Paddy milled/yr	~850 LMT
Rice bran oil	~10 LMT
Domestic edible oil	122 LMT
Edible oil imported	168 LMT
Oilseeds grown	40.99 MT

The milling that makes free PDS rice already throws off the bran for domestic oil.

STATE EXPENDITURE

CENTRAL FOOD SUBSIDY FY25

₹2.05 L cr

free rice/wheat to ~80 cr (NFSA+PMGKAY) + ₹7,075 cr FPS margins

TN SPECIAL-PDS OIL+DAL

₹3,800 cr

palmolein at ₹25/L on ~₹65/L subsidy — **on an import** (AP/KA/TG/WB run their own)

ONORC COVERAGE

36/36

States/UTs · 20.54 cr cards · portable, deduplicated

The state already spends **₹2 lakh cr+** and runs **5.43 lakh shops** to feed 80 cr people — the demand rail is built. **ONORC** turns it into one portable, deduplicated national offtake: redirect the state *oil* subsidy from imported palmolein to **domestic RBO**, and the same rupee funds rice-mill, solvent-plant and oilseed-farm jobs instead of foreign crushers — import substitution and employment on infrastructure that already exists.

Sources: PRS Demand-for-Grants 2024-25 (Food & PD) · DFPD/PIB Year-End 2024 · TNCSC · DCMSME · SEA of India · data/pds_onorc_jobs_volumes_spend.csv

PER-STATE · WHO ALREADY SUBSIDISES OIL & PULSES IN THE RATION

The oil subsidy is real, state-run, and spent on imports

STATE	OIL IN PDS	PULSE IN PDS	CARD PRICE	SPEND ₹CR/YR
Tamil Nadu	Palmolein 1 L/card	Tur dal 1 kg	oil ₹25/L · dal ₹30/kg	3,800
Karnataka	Palm oil + Indira kit	Tur/moong 1 kg	dal ~₹38/kg	~360*
Gujarat	Edible oil + sugar	Chana + tur dal	subsidised	n/p
Andhra Pradesh	—	Tur dal 500 g	subsidised	n/p
Maharashtra	—	Chana + urad	₹35/kg	n/p
Telangana	—	Red gram dal	subsidised	n/p
Kerala (Supplyco)	Oil (below-mkt)	Pulses	ltd subsidy	n/p
Centre (NAFED)	—	5.5 LMT → 11 states	varies	6,999†

TN PALMOLEIN BOUGHT

156 lakh L

+ 20,000 MT tur dal/yr — all imported oil

STATES RUNNING OIL-PDS

4 major

STATES TAKING NAFED PULSES

11+

	TN · Karnataka · Gujarat · Kerala	pulses far more widespread than oil

Tamil Nadu is the flagship: ₹3,800 cr a year, almost entirely on *imported* palmolein — the single largest identifiable state oil subsidy a domestic-RBO switch would redirect. *Karnataka figure is dal-only (oil delivered in-kit); **n/p** = scheme confirmed on state portals but per-commodity spend not published — not zero. †NAFED chana, free to states, 2020.

Sources: TNCSC PDS scale · state civil-supplies portals · NAFED via Business Standard · press 2024–25 · data/pds_state_oil_pulse_expenditure.csv

THE FISCAL CASE · SAME RUPEE, WITH VALUE CAPTURE

₹2 lakh cr to give grain away — plus ₹2 lakh cr to import what it could make

WHAT THE SYSTEM COSTS NOW

MSP paid to ~1.3 cr farmers	₹2.32 L cr
Food subsidy (free grain, 80 cr)	₹2.05 L cr
Edible-oil imports	₹1.68 L cr
Pulse imports	₹0.31 L cr
Excess-stock carry (~330 LMT over buffer)	₹18,000 cr
Rice bran given away free	₹7,650 cr

No double-count: procurement sits *inside* the subsidy. The two clean burdens are the ₹2.05 L cr subsidy + ~₹2.0 L cr imports.

WHAT PROCESSING RECOVERS

Import substitution (processing)	₹21,087 cr
— of which capturable now	~₹11,000 cr
Byproduct monetised (bran → RBO)	₹7,650 cr
Excess-stock carry avoided	up to ₹18,000 cr
Processing-margin uplift	3–5×

Branded staple processing earns 11–12% EBITDA vs 2–4% raw trading — the money is in the processing, not the sourcing.

Nothing in the food-security guarantee is cut. Farmers still get MSP; 80 cr still get free grain. What changes: the byproducts stop being given away and the oil/pulses stop being imported — the *surplus and the byproduct are processed* instead of dumped and re-imported. A value-capture layer bolted onto a ₹4 lakh-cr/yr machine, not a new subsidy.

Sources: PIB/FCI procurement (wheat ₹62,156 cr RMS25–26) · PRS food subsidy ₹2.05 L cr · DGCI&S imports · FCI stock 738 LMT (1 May 25) · report/fiscal-rationalisation-procurement-vs-processing.md

NATIONAL BUILD-OUT · STORAGE × FARMER-AGGREGATION

The grain and the farmers are in different states — design for it

STATE	FCI STORAGE LMT	SFAC FPOS
Madhya Pradesh	200	348
Punjab	175	78
Haryana	100	119
Uttar Pradesh	47	758
Bihar	21	312
Andhra Pradesh	25	247
Rajasthan	16	225
Maharashtra	24	213

FCI+hired storage (Nov-24 bulletin, ~841 LMT all-India) · SFAC 10,000-FPO registry (3,649, Feb-25 — SFAC channel only)

THE MISMATCH = THE DESIGN RULE

- **Storage belt** (Punjab 175, MP 200, Haryana 100) holds the grain — but is thin on FPOs (Punjab 78)
- **FPO belt** (UP 758, Bihar 312, AP 247, Rajasthan 225) has the farmers — but far less storage
- **MP is the one state with both** — 200 LMT + 348 FPOs + soybean/pulses/mustard = **the natural first hub**

Siting logic: oilseed/pulse crush → FPO-dense deficit states (MP, Maharashtra, Rajasthan, AP); rice-bran → the paddy/storage belt (Punjab, Haryana, Chhattisgarh). Match each stream to the state that has its feedstock *and* the aggregation.

Sources: DFPD Foodgrain Bulletin Nov-2024 (storage) · SFAC state-wise 10,000-FPO list 24-02-2025 · data/national_fci_fpo_by_state.csv

THE VEHICLE · DON'T CREATE A PSU — REVIVE THE ONE DFPD OWNS

DFPD already owns a dormant edible-oils processing PSU

Per igod.gov.in, DFPD's org chart already holds **three** delivery bodies:

FCI
procurement · storage · the grain + bran feedstock

CWC
the warehousing network

HVOC — Hindustan Vegetable Oils Corp
a CPSE built for vegetable-oil / vanaspati processing — now dormant

India needn't *create* a food-processing PSU — it can *revive* HVOC: the corporate shell, the DFPD mandate and the edible-oil remit already exist, alongside the grain (FCI) and the storage (CWC).

BUILD IT LIKE THE SURVIVORS, NOT THE FAILURES

- **Infrastructure + processing, NOT trading** — STC→PEC→NCEL died trading naked
- **FPO-fed supply** — the AMUL/NAFED survival trait
- **Ring-fenced, contracted offtake** — PDS/ONORC oil + fortification, not the open market
- **JV/PPP capital** — HVOC brings mandate + sites; partner brings crush/refining
- **Sit on the FCI/CWC network** + the ₹7,650 cr free bran

> A revived HVOC is the **edible-oil counterpart to FCI**: FCI holds the grain, HVOC processes the oil, CWC stores, FPOs supply, ONORC distributes — anchoring the ₹1.68 lakh-cr import substitution and supplying the PDS oil ration domestically instead of buying imported palmolein.

Sources: igod.gov.in DFPD org list (FCI/CWC/HVOC) · report/psu-agri-ventures-history.md (STC/PEC/NCEL vs CWC/AMUL) · report/national-fci-fpo-food-processing-psu.md

THE MODEL · WHY GOVERNMENT-COOPERATIVE-PPP

Ring-fence the commercial arm; contract the offtake

STRUCTURE

- **PACS/FPO** aggregates paddy + bran (already the procurement centre)
- **Private partner** builds + runs RBO extraction, silica, power (the capital + know-how)
- **State** provides VGF/PMKSY grant + a fortified-rice / RBO **offtake guarantee**
- Ring-fenced from MSP/buffer policy — the COFCO lesson, not the STC/Thailand failure

WHY NOT LEAVE IT TO PRIVATE MILLERS

- 100,000 mills do basic CMR — but won't invest in bran stabilisation, RBO, silica, cogen
- The integrated **value-added** mill is the PPP white space (verified: largely untried)
- The offtake guarantee is the revenue backbone — mirrors PEG's silo-hire guarantee

> History's warning (verified): STC→PEC→NCEL failed by trading naked; CWC, AMUL, Kendriya Bhandar succeeded by infrastructure / farmer-ownership / ring-fencing. This is an *infrastructure + processing* play, not a trading arm.

Sources: PPP rice-mills · PSU-agri-ventures-history · agri-trader-playbook

ILLUSTRATIVE PILOT

One integrated complex, in an operational food park

ANCHOR: RICE MILL

+ RBO EXTRACTION

+ POHA/MURMURA LINE

₹80 L 1 TPH, 19% IRR (NCDC DPR)	**₹15–40 cr** 8–15% margin, the ₹7,544 cr anchor	**₹21 L** 10–15% margin, <3y payback
+ HUSK POWER/SILICA **cluster** energy self-sufficiency + tyre-grade silica	CAPITAL SUBSIDY **35–50%** PMKSY cluster + PMFME units	OFFTAKE SECURED BY **contract** FRK/PDS + food processors + fortification

> The proof-points already exist as verified DPRs and operational parks. The novelty is **integration + the offtake guarantee** — assembling proven units into one government-anchored complex that captures the cascade instead of gifting it.

All component economics from real project DPRs; integration is the un-built step

WHY IT MATTERS · DOUBLING FARMERS' INCOME

Real farm income is ~a third of the way to doubling

The 2016 goal: double farmers' *real* income by 2022-23. Deflated by CPI-AL (the farm household's own cost basket, MoSPI), the **Real Farm Income Index** reads:

2016-17
RFII 99

→

2021-22
RFII 131

→

Target
200 = doubling

Achieved ~5.7%/yr real vs the 10.4%/yr needed — reaching 200 slips to ~2030.

THE LEVER THE GAP NEEDS

- Crop income is only **37%** of farm income — price alone can't double it
- The other 63%: wages, livestock, **non-farm/value-addition**
- Cutting *cost* defends real income against inflation the nominal headline hides
- Exactly the byproduct-cascade + processing-margin this project builds

RFII is a monitorable metric: CPI-AL deflator refreshes monthly (MoSPI), income on each SAS/NAFIS survey.

Sources: NABARD NAFIS 2021–22 · NSSO SAS · MoSPI CPIALRL · Dalwai Committee (DFI)

THE FARM-SIDE LOOP · FUEL CHOICE

Diesel→CNG + residue: the inflation-proof income lever

FUEL SHIFT (DIESEL→CNG)

+₹5,400

/yr per 1.5-ha paddy household · 24% on tractor fuel · +3.3 RFII pts

FULL CBG LOOP (+RESIDUE)

+₹9,750

/yr · residue→CBG + FOM slurry · +5.9 RFII pts · closes 9% of the doubling gap

MILL FUEL CHOICE

₹1 vs ₹10.45

husk/CBG per kWh vs grid · ₹2.7-7.9 cr/yr on the mill's EBITDA

The loop closes: crop → residue → CBG → fuels the tractor (24% cheaper, inflation-proof) AND the mill's boiler → slurry to the field. Diesel is a volatile imported input; CBG from the farmer's own residue defends the *real* income the doubling metric measures.

CACP A2+FL · PIB CNG-tractor (PRID 1697555) · SATAT CBG · TN diesel/CNG Jul 2026

THE ONE-LINE CASE

Stop gifting the byproduct. Contract the demand. Capture the cascade.

India is the world's #1 producer of the raw byproduct and a net importer of the refined derivative — for oil, gluten, starch, protein, and Vitamin-E alike. The feedstock is procured, the subsidy is 35–50%, the parks are built, the DPRs are proven. What's missing is the integrated complex and the offtake guarantee. That is the project.

SAVINGS	BEST DPR MARGIN	BEST DPR IRR
₹21k cr/yr	**poha 10–15%**	**dal 25–31%**

Full analysis, data & sources: github.com/herrrickshaw/FCI-warehouse-MSPvsMSV · research synthesis, not investment advice